SaHV: Spectral Composition-Aware Visualization of Hyperspectral Image

Jiahui Sheng, *Graduate Student Member, IEEE*, Shu Xiang, *Graduate Student Member, IEEE*, Xiaorun Li, and Shuhan Chen, *Member, IEEE*

***Abstract*—Hyperspectral images (HSIs), as a unique type of 3-D image, have a wide range of applications across various fields. However, their multichannel characteristics make them challenging for direct visual observation by humans, leading to significant inconveniences. To address this issue, the most common approach is to select three channels to generate an RGB image for visualization. However, this method neglects the most crucial spectral information in HSI, failing to represent the spectral composition of objects. Therefore, in this study, we proposed an HSI visualization method called *SaHV*, which aims to represent the spectral composition in the visualization results. The specific process of *SaHV* is as follows. First, the proposed method used spectral unmixing to obtain the abundance maps, which stands for the spectral composition. Then the obtained abundance maps were processed by our color mapping network (CMN) to produce hue and saturation channels of HSV, in which the force-directed layout method was introduced to keep local consistency. Finally, the original HSI was used to obtain the value channel of HSV. To demonstrate the effectiveness of our method, we conducted experiments on three real-world datasets.**

***Index Terms*—Force-directed layout, hyperspectral image (HSI) visualization, spectral unmixing.**

I. Introduction

UNLIKE the visible image and the infrared image, the hyperspectral image (HSI) possesses hundreds of channels that provide rich spectral information [1], [2], [3], [4], [5]. Due to its detailed spectral information, there are many different applications of HSI, such as hyperspectral target detection [6], [7], [8], HSI classification [9], [10], [11], hyperspectral super-resolution [12], [13], [14], and unmixing [15], [16]. In addition, with the continuous development of large-model technologies and in light of the unique characteristics of hyperspectral imagery, large hyperspectral models have also made significant progress in recent years [17], [18], [19], [20]. Although the additional spectral dimensions offer significant advantages to HSIs, they also complicate the direct interpretation of HSIs for users. This is because HSIs can be viewed as 3-D cube data and the third dimension usually extends far beyond three, which poses challenges to human perception [21], [22].

To enable the observation of HSIs, some HSI visualization methods are developed, which aim to map hundreds of channels of HSI to three channels of RGB [21], [23], [24], [25], [26], [27]. However, most of these methods focus on how to find or fuse the three most representative channels, whose goal is not to reflect the spectral composition and the relationships of the objects in the visualization results [24], [25]. Although some methods, such as principal component analysis (PCA), implicitly reflect the spectral correlations between objects to some extent, the principal components of PCA do not explicitly contain physical meaning, making it impossible to reflect the composition of the spectrum. At the same time, due to factors such as atmospheric interference and lighting conditions, objects made of the similar material may exhibit different spectral characteristics (similar material, different spectra), while objects made of different materials may show similar spectra (different materials, similar spectra). These issues cannot be resolved by methods such as PCA.

To address the above issues, we used hyperspectral unmixing techniques. Hyperspectral unmixing decomposes the HSI into endmembers and their corresponding abundances, which is similar to the principal components and their corresponding coefficients obtained from PCA [15], [16]. However, PCA only processes data at a numerical level, and the principal components and coefficients obtained do not explicitly have physical meaning. In contrast, unmixing bases on the physical properties of hyperspectral data, yielding endmembers and abundances that represent the composition of the spectra of ground object. Moreover, unmixing can also greatly reducing the dimensionality of the data. Related concepts of HSIs can be found in the supplementary material.

To visualize the spectral composition of ground objects in HSIs, we develop a hyperspectral visualization method, which is named as SaHV. A schematic diagram of SaHV is shown in Fig. 1. Specifically, we first conducted unmixing on HSI to obtain the abundance maps corresponding to different endmembers. Then, the abundance maps are taken as the input of our designed color mapping network (CMN) to obtain the hue and saturation maps of HSV color space (one type of color space). In the CMN, we introduce attractive and repulsive forces between different pixels. Subsequently, the original HSI is used to obtain the value map of HSV color space. Finally, we obtained visualization result by concatenating the hue, saturation, and

Received 27 September 2025; revised 28 January 2026; accepted 23 February 2026. Date of publication 2 March 2026; date of current version 10 March 2026. This work was supported by the National Key Research and Development Program of China under Grant 2024YFF1400900. *(Jiahui Sheng and Shu Xiang contributed equally to this work.) (Corresponding author: Xiaorun Li.)*

The authors are with the College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: jiahuisheng@zju.edu.cn; 12210076@zju.edu.cn; lxr@zju.edu.cn; 11410057@zju.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2026.3669404, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2026.3669404

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

value maps. The main contributions are summarized as follows.

1) We innovatively introduced unmixing into the visualization of HSIs, leveraging the physical properties of HSIs. Through unmixing, we obtained the spectral composition of the ground objects, while effectively reducing the dimensionality of HSI, significantly minimizing the dimensional span when mapping spectra to color.
2) To capture the spectral relation between pixels in the visualization result, we incorporated attraction and repulsion forces between pixels in the CMN. This approach ensures that the color distribution of the visualization result remains consistent with the composition of the spectra in the HSI.
3) We propose a spectral composition-aware HSI visualization method, which has shown good visualization results and excellent metrics.

The rest of this article is organized as follows. Section II provides a brief introduction of the related works. In Section III, we present a detailed description of our proposed hyperspectral visualization method SaHV, including the unmixing stage and the color mapping stage. In Section IV, we conducted comparative experiments for SaHV. Section V concludes this article. Finally, we provide some additional materials in the Appendices, including the definition of the metrics and the details of the unmixing network.

II. Related Work

A. Spectral Unmixing

Due to limitations in the performance of spectrometers and imaging conditions, HSIs often have low spatial resolution, resulting in multiple substances being present within each pixel. Spectral unmixing is a commonly used technique in HSI processing, aimed at analyzing the fundamental components of the image and their corresponding proportions, referred to as endmembers and abundances. The linear mixing model (LMM) [28] assumes that the spectrum of mixed pixels is derived from the linear combination of endmembers, providing a physically meaningful and general algorithm for hyperspectral unmixing. Typically, $X \in \mathbb{R}^{L\times H\times W}$ represents the HSI, where H and W denote the height and width of the image, respectively, and L represents the number of spectral bands. Thus, the linear mixing model can be expressed as

$$X = EA + \epsilon \tag{1}$$

where $E \in \mathbb{R}^{L\times P}$ and $A \in \mathbb{R}^{P\times H\times W}$ denote the endmembers and the abundance, respectively. And P represents the number of endmembers, which corresponds to the kinds of substances present in the HSI. To reflect real-world scenarios accurately, abundances must adhere to nonnegative and sum-to-one constraints [29], [30].

In real-world scenarios, the presence of nonlinear interactions between photons, along with factors such as illumination and shadow, can lead to variations in the spectra of the same substance within HSIs. To address this issue, [29], [31], [32], [33] have modeled mixed spectra based on nonlinear interactions, providing an effective method to mitigate the impact of nonlinearity on unmixing accuracy. In addition, [34], [35], [36], [37] have explicitly modeled spectral variability from the perspective of various imaging factors affecting spectral interactions.

B. HSI Visualization

HSI visualization is an technique that can project HSI into a color image for providing a quick overview of the HSI [21], [38], [39]. In the past two decades, some works have been made in this fields [23], [24], [25], [26], [27]. Existing visualization methods can be roughly divided into two main categories: band-selection-based methods [25], [40], [41], [42], [43], [44] and transform-based methods [45], [46], [47]. Among these methods, band-selection-based methods are the most straightforward methods which seek to select three representative bands of HSI based on some indicators. Although band-selection-based methods are simple and efficient, they are typically designed from the perspective of minimizing information loss, and thus often overlook the physical attributes of HSI. To address this issue, researchers have proposed a series of transform-based and other approaches that aim to preserve the physical correlations among pixel spectra while achieving hyperspectral visualization. Among them, PCA [48] is one of the simplest visualization methods that roughly accounts for spectral correlations of pixels. In addition, methods such as color matching function-based (CMF) method [46], linear spectral weighting-based method [47], and decolorization-based method (DHV) [21] have also been proposed for better performance. Despite considering spectral correlations, these methods predominantly characterize correlation in terms of vector similarity; their visualization objective remains to depict distances among spectral vectors of pixels, rather than to reveal the spectral composition of each pixel. Therefore, we propose a spectral composition-aware hyperspectral visualization method called SaHV, which aims to provide a more physically meaningful representation of HSI.

III. Methodology

A. Overall Framework

As we previously discussed, hyperspectral visualization is a technique that maps HSIs containing hundreds of channels into three-channel color images. It can also be viewed as a dimensionality reduction task for hyperspectral data. This process is extremely challenging, as it requires reducing data with hundreds of dimensions down to three, while trying to maintain as much information as possible. In addition, we also hope to roughly and preliminarily represent the composition of the spectrum through visualization, which makes it even more challenging.

To address this, we have introduced unmixing techniques, which are based on the physical characteristics of HSIs [28]. In our method, unmixing acts as an enhanced version of PCA, decomposing the HSI into a small number of components based on their spectral composition and obtaining the coefficients (abundance maps) of each component.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

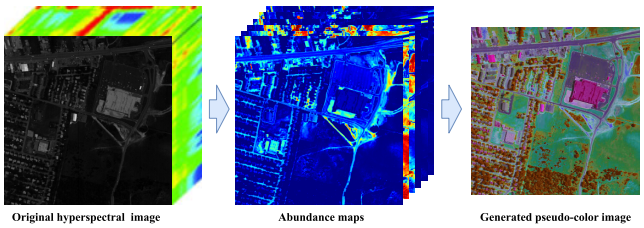


Fig. 1. **Spectral composition-aware visualization result of HSI.** We first use unmixing to convert the HSI into abundance maps, and then generate the visualization color image based on the abundance maps.

The obtained abundance maps are then fed into our CMN, where an innovative pixelwise attraction and repulsion mechanism is used. This mechanism ensures that we can fully use the entire color space while explicitly maintaining the consistency between abundance differences and color differences among pixels. The CMN will output the hue map and saturation map in HSV.

Finally, we will generate the value map based on the energy of the HSI. By integrating this value map with the previously obtained hue and saturation maps, we can derive the HSV color image of the HSI.

The overall framework of SaHV is illustrated in Fig. 2, including the unmixing stage and the color mapping stage. Next, a detailed introduction of two stages is provided.

B. Unmixing Stage

In this section, we will provide an introduction of the unmixing stage, including our shadow-aware unmixing network (SaUN). Considering the impact of shadows on spectral composition analysis, we develop the SaUN unmixing network. This network consists of an adaptive spatial aggregation encoder (ASAE), a shadow endmember-assisted decoder (SEAD), and shadow-aware endmembers, as shown in Fig. 2.

1) Adaptive Spatial Aggregation Encoder: Due to the extensive scope of remote sensing hyperspectral imaging, higher demands are placed on the computational complexity of visualization algorithms. While self-attention mechanisms are effective for feature extraction, their computational complexity increases quadratically with sequence length, making them unsuitable for visualization tasks. Thus, we propose the adaptive spatial aggregation encoder. Our approach is based on the self-attention mechanism to extract and fuse neighborhood information. In contrast to standard self-attention mechanisms, we implement unidirectional attention computations where the central pixel points to a $W \times W$ neighborhood (W is the size of windows). This strategy leverages the efficient feature extraction capabilities of attention mechanisms while avoiding information redundancy caused by bidirectional attention computations. During aggregation part, to control the contribution of neighborhood information and prevent the neglect of features from the central pixel, we use a gating mechanism that determines the importance of neighborhood information based on the dissimilarity between the central pixel and the spatial information within the neighborhood.

2) Shadow Endmember-Assisted Decoder: When capturing HSIs, varying lighting conditions such as the angle of incidence and occlusion can lead to the appearance of shadows, which may result in a generally weaker spectral intensity for objects affected by shadows, leading to the phenomenon of different spectra for the same object. Therefore, to avoid the impact of shadows on spectral composition analysis, in the proposed SEAD, we introduce auxiliary endmembers with low spectral energy (referred to as shadow endmembers) into the endmember library to model shadows in HSI. The aggregated features undergo multiple nonlinear transformations to achieve dimensionality reduction. To prevent the loss of crucial information during dimensionality reduction, gated linear units (GLU) are used. In addition, the abundances of the shadow endmember and the true endmember within a single pixel should sum up to one. Therefore, we apply a Softmax function to constrain the output low-dimensional vector. Thus, the unmixing network is capable of obtain the shadow map $\tilde{A}$ and the abundance maps A. The abundance maps A is a feature maps with P dimensions.

Notably, the unmixing network is achieved through self-supervised training. The output abundances, reconstructed with shadow-aware endmembers, are transformed back into the HSI through a reconstruction loss function. The details of the unmixing network and the training process can be found in the Appendices.

C. Color Mapping Stage

CMN is primarily responsible for deriving the hue and saturation maps of the HSV color image. In this section, we will provide a detailed explanation of the architecture of the CMN, two-branch generator, and the force-directed layout algorithms.

1) Network Architecture: The overall architecture of CMN is illustrated in Fig. 2. The CMN adopts a framework similar to an autoencoder [49], which includes a two-branch generator and a decoder, with the two-branch generator essentially functioning as a two-branch encoder. The two-branch generator further consists of a hue-branch generator and a saturation-branch generator. The hue-branch generator is responsible for producing the hue map, while the saturation-branch generator is tasked with generating the saturation map. The hue map and saturation map can be regarded as the latent variables in the autoencoder.

2) Two-Branch Generator: To compel the network to explicitly learn latent variables akin to hue and saturation, we have designed a two-branch generator. Next, we will introduce these two branches separately.

a) Hue Branch: The hue branch is specifically designed to generate the hue map, with its input being the abundance maps A, excluding the shadow map $\tilde{A}$. Once the abundance maps are fed into this branch, it is processed through the HueMapping encoder, which consists of convolutional layers, ReLU [50], batch normalization (BN) layers [51], and Tanh layers. After passing through the feature extraction encoder, the abundance maps will be mapped to 2-D features.

Since the hues in the HSV color space are typically represented as colors uniformly distributed along a circle, the features obtained from the HueMapping encoder are then be normalized to circumference of the circle. This allows us

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

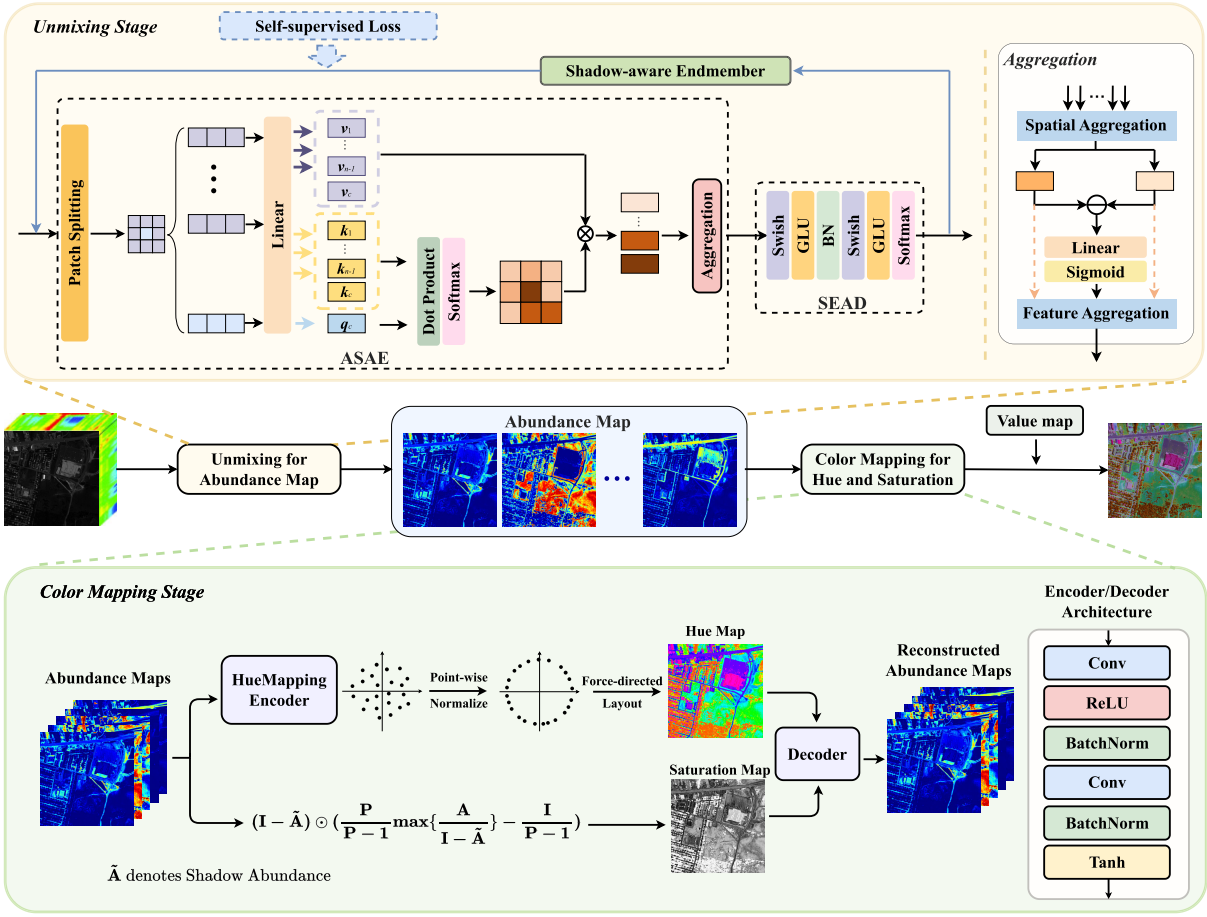


Fig. 2. Overview of the framework. The overall framework is roughly divided into two stages: unmixing stage and color mapping stage. In the unmixing stage, the HSI is unmixed by the SaUN network to get its abundance maps (including the shadow map). In the color mapping stage, the obtained abundance maps are sent to the CMN to get hue and saturation maps, and we will directly compute the value map from the original HSI.

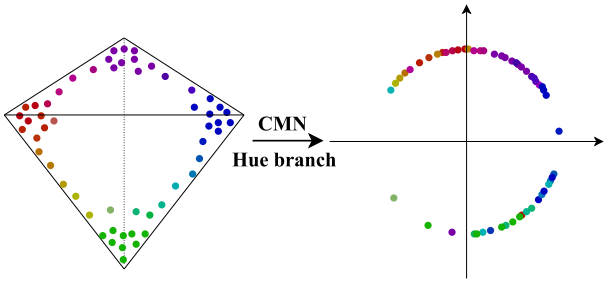


Fig. 3. Schematic illustrating the functionality of the hue branch without the force-directed layout strategy. (Left image) represents the distribution of pixels within the simplex defined by the abundance maps, while (right image) illustrates the distribution of pixels in the hue space. Closer colors indicate more similar abundances.

to obtain latent features that align with the format of hue. As shown in Fig. 3, abundance vectors typically lie on the boundary of a high-dimensional simplex (usually of dimension >2). This set is not homeomorphic to the hue circle; therefore, there is no homeomorphism that establishes a one-to-one correspondence between abundance and hue.

b) Saturation Branch: Unlike the hue branch, the saturation branch does not include any neural network blocks. Instead, we derive the corresponding saturation map directly using our saturation formula, which is as follows:

$$S = (I - \tilde{A}) \odot \left(\frac{P}{P-1} \max \left\{ \frac{A}{I - \tilde{A}} \right\} - \frac{I}{P-1} \right) \tag{2}$$

where A is the abundance maps, $\tilde{A}$ is the shadow map, P is the number of abundance maps, $\odot$ represents the Hadamard product, and I denotes the matrix of ones that matches the shape of $\tilde{A}$, and the max operation represents taking the maximum value along the channel dimension of the abundance maps.

According to the saturation formula, pixels with a more uniform distribution of abundance values will exhibit lower saturation. When a pixel is composed equally of all the endmembers, its saturation will be zero. This is because we interpret saturation as a measure of the certainty regarding the spectral composition of that pixel. In unmixing, a more uniform abundance distribution typically indicates that the spectral composition of the pixel is uncertain, thereby justifying a reduction in the saturation of that pixel to dilute its color (in the HSV color space, lower saturation corresponds to a lighter color). In addition, we also consider the influence of the shadow; as the value in the shadow map increases, the uncertainty of the unmixing result grows, resulting in lower saturation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

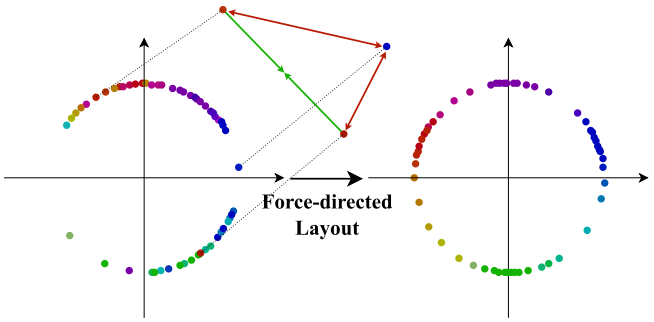


Fig. 4. Schematic of the force-directed layout strategy. Attractive forces draw the hues of similar abundance points closer together, while repulsive forces push the hues of dissimilar abundance points apart. We can find the utilization rate of hue improved by using the force-directed layout strategy.

3) Force-Directed Layout: Relying solely on the hue branch generator to obtain the hue map may lead to insufficient color utilization and inconsistency between hue and abundance. In the visualization results, insufficient color utilization manifests as the visualized image being predominantly biased toward a single color, while inconsistency between hue and abundance means that similar abundance has different hue but different abundance has similar hue, as shown in Fig. 3. Compared with other methods that may implicitly reflect the consistency between hue and abundance, we have explicitly ensured this consistency through force-directed layout.

To address this issue, inspired by the force-directed layout algorithm [52], [53], [54], we introduced attractive and repulsive forces between the points on the hue circle. As illustrated in Fig. 4, attraction force causes points with similar abundance to move closer together on the hue circle, while repulsion force can keep the distance between points with significant abundance differences, which can also increase the utilization rate of colors.

To maintain consistency between hue and abundance, first obtain the relationships among the various pixels in the abundance space. To achieve this, we constructed local relationships in the abundance space using a nearest neighbors graph, specifically a weighted K-nearest neighbors graph. Let $A = a_1, \ldots, a_N$ represents the abundance maps. For each a_i, we calculate the weights between it and its K nearest neighbors. Then, we can obtain a weighted directed graph $G = (V, E, \omega)$. The $\omega_{i,j}$ is the weights of the directed edge from a_i to a_j, which is defined as following:

$$\omega_{i,j} = \exp\left(\min\left\{d\left(a_i, a_{ik}\right)\right\} - d\left(a_i, a_j\right)\right) \tag{3}$$

where a_{ik} represents the k nearest neighbors of a_i, and $d(a_i, a_j)$ represents the Euclidean distance between a_i and a_j.

The forces act reciprocally; however, the directed graph we obtain is not symmetric. Therefore, we process the adjacency matrix of the graph as follows to make it symmetric:

$$B = \Lambda + \Lambda^T \tag{4}$$

where Λ represents the adjacency matrix.

a) Attraction Force: The attraction force between vertices i and j is determined by

$$F_a(i, j) = \left(1 + \left\|h_i - h_j\right\|_2\right)^{2\omega_{i,j}} \tag{5}$$

where h_i and h_j represent the value of vertices i and j in hue space, respectively. $\omega_{i,j}$ is the value at the corresponding position in matrix B. Thus, the attraction loss can be described as

$$L_a = \sum_{i=1}^{N} \sum_{j \in \text{K neighbors}} F_a(i, j). \tag{6}$$

b) Repulsion Force: Similar to the attraction force, we do not compute the repulsion force between all the points to reduce the computing cost. Instead, we randomly select K points for each point to calculate the repulsion force, which is defined as follows:

$$F_r(i, j) = \begin{cases} \dfrac{\left\|a_i - a_j\right\|_2}{\left(1 + \left\|h_i - h_j\right\|_2\right)^2} & \text{if } \left\|a_i - a_j\right\|_2 > d_{\text{ref}} \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

where d_{ref} is given by

$$d_{\text{ref}} = \frac{1}{P}\sqrt{1 - \frac{1}{P}} \tag{8}$$

where P is the number of channel of abundance maps.

The repulsion loss can be described as

$$L_r = \sum_{i=1}^{N} \sum_{j \in \text{K samples}} F_r(i, j). \tag{9}$$

Algorithm 1 Calculate Attraction and Repulsion Loss

Input: Abundance maps A; symmetric adjacency matrix B; hue map of HSI H; initial nearest neighbor number K.
Output: Attraction Loss L_a and Repulsion Loss L_r.
Initiate: $L_a \leftarrow 0$, $L_r \leftarrow 0$
for $i = 1$ to N **do**
  for $j \in$ K neighbors **do**
    $F_a(i, j) \leftarrow (1 + \|h_i - h_j\|_2)^{2\omega_{i,j}}$
    $L_a \leftarrow L_a + F_a(i, j)$
  end for
  for $j \in$ K samples **do**
    if $\|a_i - a_j\|_2 > d_{\text{ref}}$ **then**
      $F_r(i, j) \leftarrow \frac{\|a_i - a_j\|_2}{(1 + \|h_i - h_j\|_2)^2}$
    else
      $F_r(i, j) \leftarrow 0$
    end if
    $L_r \leftarrow L_r + F_r(i, j)$
  end for
end for

The specific steps for calculating the attraction and repulsion loss are detailed in Algorithm 1.

In summary, the loss function for the CMN can be formulated as

$$L = \lambda L_{rec} + L_a + L_r \tag{10}$$

where L_{rec} represents the reconstruction loss, which is defined as the mean-squared error (mse) between the original abundance maps and the reconstructed abundance maps. λ is the weighting coefficient for the reconstruction loss.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE METRICS OF DIFFERENT VISUALIZATION METHODS. THE BEST VALUES ARE HIGHLIGHTED

Methods	HYDICE Urban				Salinas				Botswana			
	SF	SD	AG	EN	SF	SD	AG	EN	SF	SD	AG	EN
BF	48.76	22.09	11.00	6.14	17.56	7.21	1.24	4.42	27.69	14.01	5.03	5.34
PCA	40.96	16.84	8.96	5.70	42.50	16.94	2.90	5.29	31.97	14.06	7.04	5.16
SPCA	47.47	21.31	11.42	6.15	64.51	28.19	4.17	6.07	34.81	18.06	7.50	5.76
SMI-BS	57.18	25.76	14.98	6.50	27.02	10.35	1.81	4.68	33.68	18.34	8.63	5.49
DHV	67.45	29.76	16.22	6.65	70.88	31.79	5.53	**6.57**	46.46	26.76	10.68	6.55
SaHV(ours)	**97.09**	**41.60**	**21.92**	**7.18**	**84.87**	**36.85**	**6.70**	6.56	**79.09**	**37.48**	**19.86**	**6.80**

D. Value Map Generate

In Section III-C, we discussed how to obtain the hue and saturation maps in the HSV color space from the CMN. In this section, we will explain how to derive the remaining value map.

Unlike the hue and saturation maps, which are intended to reflect spectral composition, we aim for the value map to represent the energy differences at various locations. Specifically, we first sum all the channels of the HSI, normalize the summed energy to $[0, 1]$, and then apply a gamma correction so that the value component better matches human visual perception, which can be written as

$$S_i = \sum_{l=1}^{L} X_{i,l} \tag{11}$$

$$V_i = \tilde{S}_i^{1/\gamma} \tag{12}$$

where $X_{i,l}$ represents the value of pixel i in the lth spectral band, $\tilde{S}_i$ is the normalized summed energy, and γ (gamma) is the gamma parameter used for gamma correction. In our experiments, we set γ to 2.2.

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: To validate the effectiveness of SaHV, we conducted experiments on three hyperspectral datasets: HYDICE Urban ($307 \times 307 \times 162$), Salinas ($512 \times 217 \times 204$), and Botswana ($1476 \times 256 \times 145$).

2) Metrics: To quantify the quality of the visualization results, we selected four metrics for evaluation: separability of features (SF) [55], standard deviation (SD) [56], average gradient (AG) [40], and entropy (EN) [40]. The detailed definitions of these metrics can be found in the Appendices.

3) Competing Methods: To demonstrate the effectiveness of SaHV, we compared it with five widely used visualization methods: the bilateral filtering (BF) method [24], the PCA method [48], the SPCA method [45], the DHV method [21], and the SMI-BS method [44].

4) Implementation Details: The initialization of endmembers in the unmixing network is obtained using the vertex component analysis method [57]. To showcase the robustness of our proposed SaHV method, we fixed the number of endmembers P at 5 and the parameter K at 10 across all the datasets. All the experiments are conducted on a Windows 11 computer with 13th Gen Intel Core i7-13700K CPU and NVIDIA GeForce RTX 4060Ti GPU.

B. Visualization Results

The visualization results and corresponding metrics of different methods are shown in Fig. 5 and Table I, respectively. We can observe that the colors in SaHV's visualization results are significantly richer than those of other method, which enhances the images clarity and improve the utilization rate of color. In addition to the richer colors, the transitions between spectra are also effectively represented in SaHV's visualization results. For instance, in the case of the HYDICE Urban dataset, we can see that the color transition between grass and soil in the SaHV's visualization image is much smoother. Moreover, Table I demonstrates that SaHV outperforms the other comparison methods across all the four metrics. Notably, in terms of SFs, SaHV significantly surpasses the other methods. This is because SF, to some extent, describes the utilization rate of color in images, which aligns well with the rich colors observed in SaHV's visualization results.

C. Ablation Study

We conducted ablation experiments on the shadow-assisted unmixing module of SaHV and the force-directed layout by, respectively, removing the corresponding modules from the SaHV. The ablation experiment on the shadow-assisted unmixing module was conducted by replacing the shadow endmember-assisted decoder with a vanilla decoder.

Since the metrics used above primarily describe overall color and brightness differences in the images, they cannot effectively capture the contributions of the two modules. Therefore, in this section, we will focus on illustrating their impact through the visualization results. From Fig. 6, we can see that the presence or absence of the shadow endmember has a minimal impact on the overall visualization results. This is reasonable, as the shadow endmember mainly addresses a limited number of shadowed areas, such as the shaded sides of buildings, and thus does not affect the overall outcome significantly. Moreover, as shown in Fig. 7, the use of the shadow endmember ensures that the hues of the shaded and sunlit sides of rooftops remain consistent. Although introducing the shadow endmember improves the hue accuracy in a small number of shadowed regions, its overall impact remains limited, especially when shadows are scarce in the HSI or when other endmembers with extremely low energy are present.

Unlike the shadow endmember, which only affects a small portion of the image, the force-directed layout impacts the overall color and details of the visualization results. In Fig. 6,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

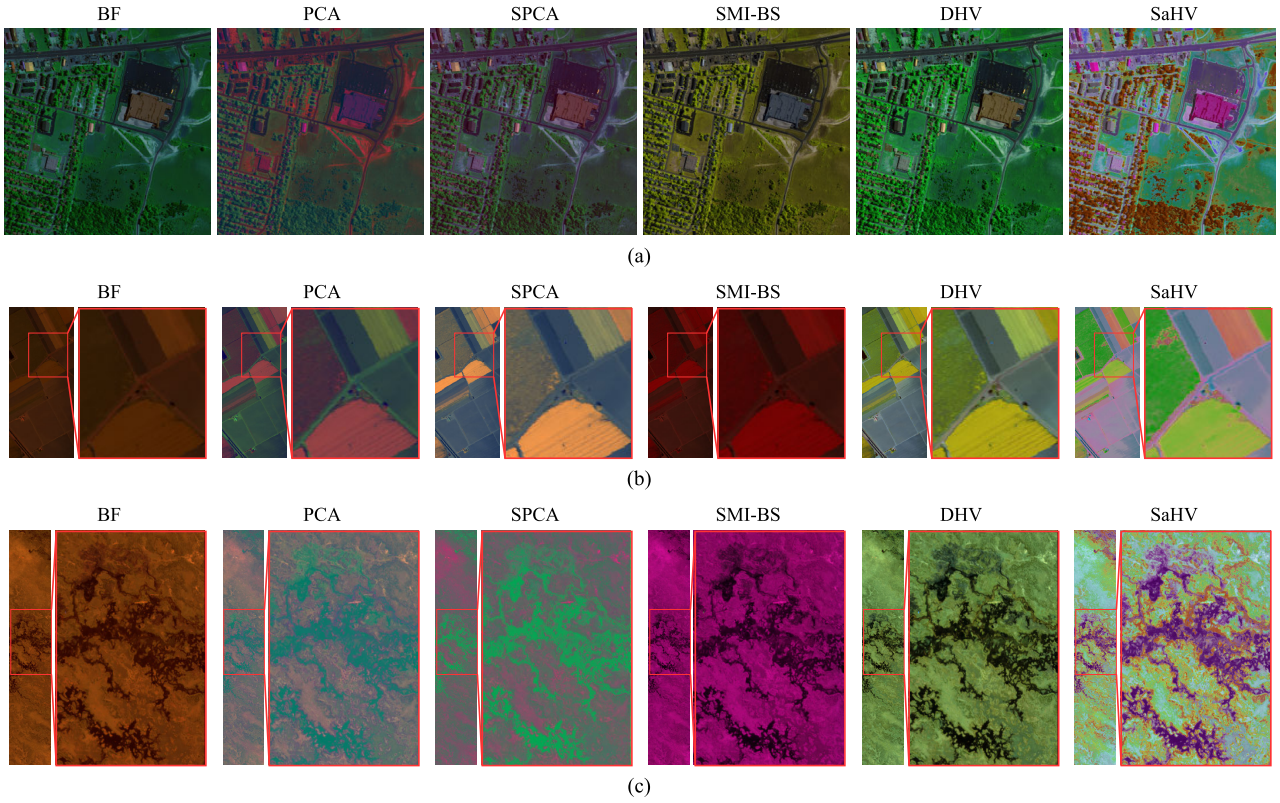


Fig. 5. Visualization results of all the competing methods on the three datasets. (a) Visualization results of different methods on the HYDICE Urban dataset. (b) Visualization results of different methods on the Salinas dataset. (c) Visualization results of different methods on the Botswana dataset.

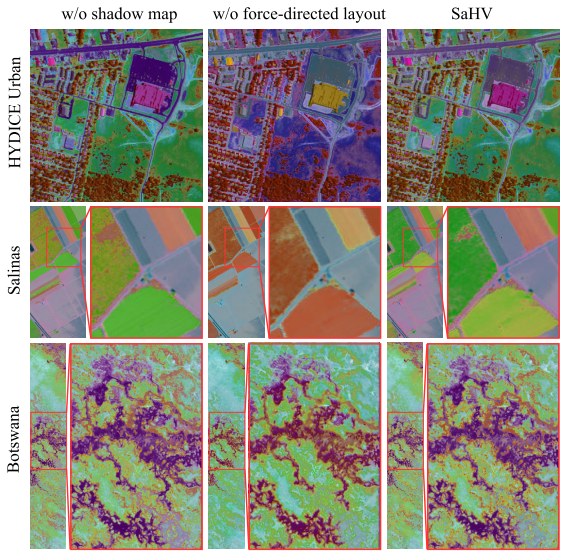


Fig. 6. Visualization results of ablation experiments.

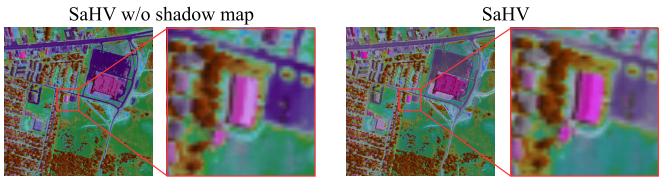


Fig. 7. Rooftop example of the HYDICE Urban dataset.

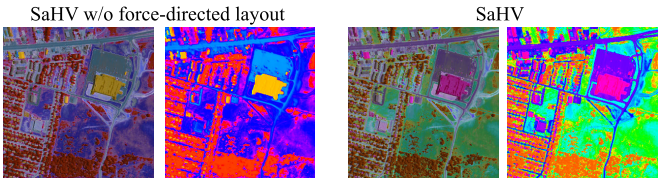


Fig. 8. Visualization results and their hue maps of force-directed layout ablation experiment on the HYDICE Urban dataset.

the visualization results of SaHV without the force-directed layout exhibit an imbalanced hue, with the HYDICE Urban dataset appearing overall darker, the lower region of the Salinas dataset lacking richness in hue, and the Botswana dataset showing a lack of detail. This imbalance is even more apparent in the hue maps, which exhibits a low color utilization rate, as shown in Fig. 8.

V. Conclusion

In this article, we propose a hyperspectral visualization method that aims to represent spectral composition in the generated color images. SaHV comprises two main stages: unmixing and color mapping. The unmixing stage performs dimensionality reduction on the HSI while obtain the spectral composition, and the color mapping stage transfer the abundance maps into the HSV color space, establishing a connection between spectral composition and color. Experiments demonstrate the effectiveness of SaHV.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

APPENDIX A
DEFINITION OF THE METRICS

The four metrics—SFs, SD, AG, and EN—are commonly used indicators in the hyperspectral visualization task. Next, the definitions of these four metrics will be introduced.

A. Separability of Features

SF refers to the degree to which distinct pixels are mapped to distinguishable colors. The larger the value of feature separability, the greater the average difference between two pixels in the color space. Generally speaking, the larger the feature separability, the better. The SF is defined as follows:

$$\mathrm{SF} = \frac{1}{(N-1)^2} \sum_{x_i \neq y_I} d\left(x_i, y_i\right) \tag{13}$$

where N is the number of the pixels, and $d(x_i, y_i)$ represents the Euclidean distance between the pixels x_i and y_i in the color space.

B. Standard Deviation

SD is used to evaluate the contrast of the visualized image, and generally, the larger its value, the better. The definition of SD is as follows:

$$\mathrm{SD} = \left(\frac{1}{(N-1)} \sum_{i=1}^{N} (x_i - \bar{x}) \right)^{\frac{1}{2}} \tag{14}$$

where $\bar{x}$ denotes the mean value of the pixels in the visualization results.

C. Average Gradient

AG measures the sharpness of an image by assessing its gradient values. The definition of AG is as follows:

$$\mathrm{AG} = \frac{1}{(N)} \sum_{x} \sum_{y} \left(\sqrt{I_x^2 + I_y^2} \right) \tag{15}$$

where I_x and I_y represent the difference in the x-axis and y-axis directions, respectively. The AG value in this article is the average of the AG values across the three channels.

D. Entropy

EN indicates the degree of information preservation in the visualized image; the higher the EN value, the greater the amount of information in the image. EN is defined as follows:

$$\mathrm{EN} = -\sum_{i} p_i \ \ln \ p_i. \tag{16}$$

APPENDIX B
DETAILS OF THE UNMIXING NETWORK

A. Implementation Details

In Section III-B of the main text, we briefly outline the overall concept of the proposed unmixing module. Thus, we provide a more detailed explanation of the implementation process for the unmixing module in the supplementary material.

The proposed shadow-aware unmixing network addresses the challenges of large-scale remote sensing images by leveraging a local attention mechanism to capture valuable information. Since unmixing is a per-pixel task, we use a $W \times W$ neighborhood around the target pixel to define a patch as spatial auxiliary information. For each pixel within a patch, we perform the projection operations of the attention mechanism, projecting the data into query, key, and value spaces. Notably, we focus exclusively on information relevant to the central pixel, i.e., the target pixel for unmixing, which is the only one projected into the query space. The projection operations can be expressed as follows:

$$Q_c = x_c P_Q, K = X P_K, V = X P_V \tag{17}$$

where $X = [x_1, \ldots, x_c, \ldots, x_{w^2}]$ represents the pixels in every patch and x_c denotes the target pixel. P_Q, P_K, and P_V are the projection matrices.

The adaptive spatial aggregation encoder (ASAE) is implemented based on a self-attention mechanism. First, correlation-based weighting is applied to extract information relevant to the target pixel, enabling the aggregation of spatial information within its neighborhood

$$\mathbf{t} = \mathrm{Softmax}\left(\frac{Q_c K^T}{\sqrt{d}} \right). \tag{18}$$

In the aggregation part, spatial aggregation can be represented by the following formula:

$$x_{spa} = \sum_{i=1, t_i \in \mathbf{t}}^{W^2} t_i \cdot v_i \tag{19}$$

$$x_c' = t_c \cdot v_c. \tag{20}$$

In x_{spa}, neighborhood spatial information is aggregated, while x_c' represents the feature of the target pixel that takes into account the neighborhood information. After that, a gating mechanism is designed to facilitate the aggregation of target pixel features with spatial information. The gating network f_{gate}, composed of a linear layer and the Sigmoid function, converts the differences between the spatial information and target pixel features into weighting factors, which control the contribution of spatial information during aggregation

$$m = f_{\mathrm{gate}}\left(x_{spa} - x_c' | \Theta_g\right). \tag{21}$$

Thus, under the adjustment of the gating factors, the module achieves effective feature aggregation

$$x_c'' = x_c' + m \cdot x_{spa}. \tag{22}$$

By concatenating all the aggregated features, we obtain the feature X_c'' of the input image. The shadow endmember-assisted decoder (SEAD) analyzes the abundance of each endmember in the image based on the aggregated features. Unlike other models, we introduce an additional shadow endmember, representing the fundamental spectral composition of shadowed regions caused by imaging conditions

$$A, \widetilde{A} = \mathcal{D}_{\mathrm{SEA}}\left(X_c''\right) \tag{23}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

where A and $\widetilde{A}$ represent the abundance maps and shadow abundance map, respectively. The decoder $\mathcal{D}_{\text{SEA}}$ is implemented using several nonlinear transformations. To prevent essential information loss during feature dimension reduction, we replaced ordinary linear layers with gated linear units (GLUs). GLUs introduce an additional gating mechanism on top of the linear transformation to control the contribution of each dimension, enabling more precise control of information flow. Specifically, the formula for GLU can be expressed as

$$\text{GLU}(x) = (Wx + b) \odot \text{Sigmoid}(Vx + c) \tag{24}$$

where $\odot$ represents the Hadamard product.

Notably, we impose a sum-to-one constraint on both A and $\widetilde{A}$ using the Softmax function, effectively preventing abnormal abundance estimations in cases where spectral values are reduced due to shadow effects.

To facilitate subsequent network training, we introduce shadow-aware endmembers to reconstruct the input HSI. Specifically, we apply a linear layer without bias to map low-dimensional abundances to high-dimensional HSIs. Thus, the weights of the linear layer serve the role of endmembers in unmixing and can be viewed as a spectral dictionary of fundamental constituents in the image. In Appendix B-B, we will present the details of the loss function.

B. Loss Functions

The unmixing network is trained based on a self-supervised manner. Leveraging the spectral unmixing model from Section II-A, the network's outputs of endmembers, abundances, and shadow map can reconstruct the HSI tensor. Therefore, the network's loss function first considers the Euclidean distance as the reconstruction loss to estimate the model's reconstruction error. The Euclidean distance can pixelwise estimate the differences between the reconstructed spectra and the original image, reflecting the overall reconstruction level

$$\mathcal{L}_{\text{mse}} = \frac{1}{N}\sum_{i=1}^{N}\left(\widehat{\mathbf{y}}_i - \mathbf{y}_i\right)^2. \tag{25}$$

In addition, considering HSIs contain high-dimensional spectral information, the shape of the spectra can reflect detailed information of various land covers and characteristic bands. Hence, we introduce spectral angle distance to evaluate the reconstruction from the perspective of spectral shape. The specific formula is as follows:

$$\mathcal{L}_{\text{SAD}} = \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\langle \mathbf{y}_i, \widehat{\mathbf{y}}_i\rangle}{\|\mathbf{y}_i\|_2 \left\|\widehat{\mathbf{y}}_i\right\|_2}\right) \tag{26}$$

$$\mathcal{L}_{\text{re}} = \alpha\mathcal{L}_{\text{mse}} + \beta\mathcal{L}_{\text{SAD}} \tag{27}$$

where $\widehat{\mathbf{y}}_i$ and $\mathbf{y}_i$ indicate the ith reconstructed spectrun and input spectrum, respectively. Weighting factors α and β are used to control the contribution of these loss functions.

Furthermore, the network imposes sparse constraints on the abundance and shadow maps. HSIs reflect the spatial distribution of actual objects, so it is rare for endmembers to be widely distributed across all the pixels; in other words, their corresponding abundance maps exhibit sparsity. Similarly, shadow pixels reflect shadows in the image due to imaging angles and occlusions. Therefore, we believe that shadow map should also exhibit sparsity. Due to the effectiveness of the l1 norm in imposing sparse constraints on matrices while maintaining convexity, we choose to apply it to impose sparse constraints on the abundance and shadow maps. This can be represented in formulas as follows:

$$\mathcal{L}_{\text{sparsity}} = \gamma\sum_{i=1}^{P}|a_i| + \delta\sum_{j=1}^{L}|\tilde{a}_j| \tag{28}$$

where a_i and $\tilde{a}_j$ represent the abundance matrix of the ith endmember and shadow matrix of the jth spectral band, respectively.

Therefore, the loss function of the unmixing network can be summarized as a reconstruction term and a regularization term. The network achieves self-supervised training based on this loss function, and the overall loss function can be expressed as

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{re}} + \mathcal{L}_{\text{sparsity}}. \tag{29}$$

C. Training Details

After constructing the network with the aforementioned details, we trained the network using the Adam optimizer. The initial learning rate was set to $1e^{-3}$. To prevent the learning rate from becoming too large in the later stages of training, which could hinder obtaining the global optimal solution, we decayed the learning rate to 0.9 of its original value every 20 epochs. In addition, the local receptive field size W in the network was set to 3. The Swish activation function was adopted to train the network more efficiently.

REFERENCES

[1] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[2] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[3] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[4] Z. Zhao, P. Wang, P. Zhang, and Z. Lv, "Sample augmentation and balance approach for improving classification performance with high-resolution remote sensed image," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[5] Z. Lv, S. Cheng, L. Xie, J. Li, and M. Zhao, "A graph contrastive learning network for change detection with heterogeneous remote sensing images," *Pattern Recognit.*, vol. 172, Apr. 2026, Art. no. 112394.

[6] X. Yang, B. Tu, Q. Li, J. Li, and A. Plaza, "Graph evolution-based vertex extraction for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 1–15, Dec. 2024.

[7] J. Lian, L. Wang, H. Sun, and H. Huang, "GT-HAD: Gated transformer for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 1–15, Feb. 2025.

[8] Y. Wang, X. Chen, E. Zhao, and M. Song, "Self-supervised spectral-level contrastive learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510515.

[9] F. Ullah, I. Ullah, R. U. Khan, S. Khan, K. Khan, and G. Pau, "Conventional to deep ensemble methods for hyperspectral image classification: A comprehensive survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3878–3916, 2024.

[10] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

[11] L. Scheibenreif, M. Mommert, and D. Borth, "Masked vision transformers for hyperspectral image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2166–2176.

[12] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[13] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[14] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[15] R. A. Borsoi, T. Imbiriba, and P. Closas, "Dynamical hyperspectral unmixing with variational recurrent neural networks," *IEEE Trans. Image Process.*, vol. 32, pp. 2279–2294, 2023.

[16] Y. Duan, X. Xu, T. Li, B. Pan, and Z. Shi, "UnDAT: Double-aware transformer for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522012.

[17] D. Wang et al., "HyperSIGMA: Hyperspectral intelligence comprehension foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 8, pp. 6427–6444, Aug. 2025.

[18] J. Li et al., "HyperFree: A channel-adaptive and tuning-free foundation model for hyperspectral remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 23048–23058.

[19] F. Luo, X. Chen, C. Fu, T. Guo, and B. Du, "HDiff-HIR: Hierarchically conditional diffusion model for hyperspectral image reconstruction," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 1, pp. 777–791, Jan. 2026.

[20] F. Luo, X. Chen, T. Guo, X. Gong, L. Zhang, and C. Zhu, "DSMT: Dual-stage multiscale transformer for hyperspectral snapshot compressive imaging," *IEEE Trans. Image Process.*, vol. 34, pp. 2473–2486, 2025.

[21] X. Kang, P. Duan, S. Li, and J. A. Benediktsson, "Decolorization-based hyperspectral image visualization," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4346–4360, Aug. 2018.

[22] Y. Dong, B. Du, L. Zhang, and L. Zhang, "Dimensionality reduction and classification of hyperspectral images using ensemble discriminative local metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2509–2524, May 2017.

[23] H. Su, Q. Du, and P. Du, "Hyperspectral image visualization using band selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2647–2658, Jun. 2014.

[24] K. Kotwal and S. Chaudhuri, "Visualization of hyperspectral images using bilateral filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 5, pp. 2308–2316, May 2010.

[25] B. Demir, A. Celebi, and S. Erturk, "A low-complexity approach for the color display of hyperspectral remote-sensing images using one-bit-transform-based band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 1, pp. 97–105, Jan. 2009.

[26] R. Tang, H. Liu, J. Wei, and W. Tang, "Supervised learning with convolutional neural networks for hyperspectral visualization," *Remote Sens. Lett.*, vol. 11, no. 4, pp. 363–372, Apr. 2020.

[27] S. L. Moan, "Hyperspectral band clustering for visualisation," in *Proc. 13th Workshop Hyperspectral Imag. Signal Processing: Evol. Remote Sens. (WHISPERS)*, Oct. 2023, pp. 1–5.

[28] C.-I Chang, S.-S. Chiang, J. A. Smith, and I. W. Ginsberg, "Linear spectral random mixture analysis for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 2, pp. 375–392, Feb. 2002.

[29] C. C. Borel and S. A. W. Gerstl, "Nonlinear spectral mixing models for vegetative and soil surfaces," *Remote Sens. Environ.*, vol. 47, no. 3, pp. 403–416, Mar. 1994.

[30] L. Drumetz, M.-A. Veganzones, S. Henrot, R. Phlypo, J. Chanussot, and C. Jutten, "Blind hyperspectral unmixing using an extended linear mixing model to address spectral variability," *IEEE Trans. Image Process.*, vol. 25, no. 8, pp. 3890–3905, Aug. 2016.

[31] W. Fan, B. Hu, J. Miller, and M. Li, "Comparative study between a new nonlinear model and common linear model for analysing laboratory simulated-forest hyperspectral data," *Int. J. Remote Sens.*, vol. 30, no. 11, pp. 2951–2962, Jun. 2009.

[32] A. Halimi, Y. Altmann, N. Dobigeon, and J.-Y. Tourneret, "Nonlinear unmixing of hyperspectral images using a generalized bilinear model," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4153–4162, Nov. 2011.

[33] R. Heylen and P. Scheunders, "A multilinear mixing model for nonlinear spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 240–251, Jan. 2016.

[34] O. Eches, N. Dobigeon, C. Mailhes, and J.-Y. Tourneret, "Bayesian estimation of linear mixtures using the normal compositional model. Application to hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 19, no. 6, pp. 1403–1413, Jun. 2010.

[35] B. Somers, G. P. Asner, L. Tits, and P. Coppin, "Endmember variability in spectral mixture analysis: A review," *Remote Sens. Environ.*, vol. 115, no. 7, pp. 1603–1616, Jul. 2011.

[36] P.-A. Thouvenin, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral unmixing with spectral variability using a perturbed linear mixing model," *IEEE Trans. Signal Process.*, vol. 64, no. 2, pp. 525–538, Jan. 2016.

[37] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[38] P. Duan, X. Kang, S. Li, and P. Ghamisi, "Multichannel pulse-coupled neural network-based hyperspectral image visualization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2444–2456, Apr. 2020.

[39] X. Kang, P. Duan, and S. Li, "Hyperspectral image visualization with edge-preserving filtering and principal component analysis," *Inf. Fusion*, vol. 57, pp. 130–143, May 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253518303051

[40] Y. Yuan, G. Zhu, and Q. Wang, "Hyperspectral band selection by multitask sparsity pursuit," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 631–644, Feb. 2015.

[41] Y. Yuan, J. Lin, and Q. Wang, "Dual-clustering-based hyperspectral band selection by contextual analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1431–1445, Mar. 2016.

[42] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[43] S. Le Moan, A. Mansouri, Y. Voisin, and J. Y. Hardeberg, "A constrained band selection method based on information measures for spectral image color visualization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 5104–5115, Dec. 2011.

[44] C.-I. Chang, Y.-M. Kuo, S. Chen, C.-C. Liang, K. Y. Ma, and P. F. Hu, "Self-mutual information-based band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5979–5997, Jul. 2021.

[45] V. Tsagaris, V. Anastassopoulos, and G. A. Lampropoulos, "Fusion of hyperspectral data using segmented PCT for color representation and classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 10, pp. 2365–2375, Oct. 2005.

[46] M. Magnusson, J. Sigurdsson, S. E. Armansson, M. O. Ulfarsson, H. Deborah, and J. R. Sveinsson, "Creating RGB images from hyperspectral images using a color matching function," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 2045–2048.

[47] N. P. Jacobson, M. R. Gupta, and J. B. Cole, "Linear fusion of image sets for display," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3277–3288, Oct. 2007.

[48] X. Jia and J. A. Richards, "Segmented principal components transformation for efficient hyperspectral remote-sensing image display and classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 1, pp. 538–542, Jan. 1999.

[49] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, Jul. 2006. [Online]. Available: https://www.science.org/doi/abs/10.1126/science.1127647

[50] G. E. Hinton, S. Osindero, and Y.-W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, Jul. 2006.

[51] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*.

[52] T. M. J. Fruchterman and E. M. Reingold, "Graph drawing by force-directed placement," *Softw., Pract. Exper.*, vol. 21, no. 11, pp. 1129–1164, 1991.

[53] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform manifold approximation and projection for dimension reduction," 2018, *arXiv:1802.03426*.

[54] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, 2008. [Online]. Available: http://jmlr.org/papers/v9/vandermaaten08a.html

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.

[55] D. Liao, Y. Qian, J. Zhou, and Y. Y. Tang, "A manifold alignment approach for hyperspectral image visualization with natural color," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3151–3162, Jun. 2016.
[56] C.-I. Chang and Q. Du, "Interference and noise-adjusted principal components analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 5, pp. 2387–2396, Sep. 1999.
[57] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.



Jiahui Sheng (Graduate Student Member, IEEE) received the B.S. degree from Zhejiang University, Hangzhou, China, in 2023, where he is currently pursuing the Ph.D. degree in control theory and control engineering.

His research interests include hyperspectral anomaly detection, image processing, and pattern recognition.



Shu Xiang (Graduate Student Member, IEEE) received the B.S. degree from Zhejiang University, Hangzhou, China, in 2022, where he is currently pursuing the Ph.D. degree in control theory and control engineering.

His research interests include hyperspectral unmixing, image processing, and pattern recognition.



Xiaorun Li received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1992, and the M.S. and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 1995 and 2008, respectively.

Since 1995, he has been with Zhejiang University, where he is currently a Professor with the College of Electrical Engineering. His research interests include hyperspectral image processing, signal and image processing, and pattern recognition.



Shuhan Chen (Member, IEEE) received the B.S. degree from Ludong University, Yantai, China, in 2011, the M.S. degree from Liaoning Technical University, Huludao, China, in 2014, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 2020.

She did research as a Visiting Faculty Assistant at the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, from 2018 to 2020. She did a Post-Doctoral position at the Department of Control Science and Engineering, Zhejiang University. She is currently an Assistant Professor at the Department of Control Science and Engineering, Zhejiang University. Her research interests include image registration, hyperspectral image processing, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:00:26 UTC from IEEE Xplore. Restrictions apply.